

boltex mfg. co., l.p.

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MATERIAL TEST REPORT - EN 10204:2004 Type 3.1

S O L D	EDGEN MURRAY CORPORATION	S H I P	PIPELINE EQUIPMENT
	P.O.BOX 84160		8403 SOUTH 89TH WEST AVENUE
	BATON ROUGE, LA 70884		TULSA OK 74131
			PO# 74769-03

ITEMS

Ln#	Qty	Description	Matl Spec	Ht Cd	Mill Heat Number	CE
002	1	12" 900 RF WN XH MSS A694 F52	A694N-16/MSS-SP-44-19	1F536N	NU4000018815	0.41
002	3	12" 900 RF WN XH MSS A694 F52	A694N-16/MSS-SP-44-19	1R231NM	NU4000034437	0.40
003	2	20" 900 RF WN SCH 40 MSS A694 F65	A694H-16/MSS-SP-44-19	1F569H1	NU4000018610	0.42

Ladle Chemical Properties Indicated with an Asterisk ()

CHEMICAL PROPERTIES

Ht Cd	C	Si	Mn	P	S	Cr	Al	Cu	Ni	Mo	V	Nb	Sn	N	B	Pb	Ti
1F536N	0.15	0.28	1.15	0.010	0.005	0.08	0.023	0.21	0.07	0.03	0.06	0.000	0.000	0.000	0.0002	0.000	0.003
*	0.15	0.28	1.21	0.008	0.003	0.10	0.018	0.21	0.07	0.02	0.07	0.000	0.006	0.017	0.0003	0.000	0.002
1R231NM	0.15	0.27	1.26	0.012	0.008	0.12	0.030	0.21	0.08	0.03	0.07	0.013	0.000	0.000	0.0006	0.000	0.002
*	0.14	0.27	1.21	0.008	0.002	0.12	0.028	0.20	0.09	0.03	0.07	0.012	0.006	0.019	0.0001	0.001	0.002
1F569H1	0.18	0.20	1.14	0.008	0.004	0.09	0.026	0.18	0.08	0.03	0.06	0.000	0.000	0.000	0.0002	0.000	0.002
*	0.17	0.19	1.20	0.008	0.002	0.08	0.021	0.17	0.07	0.02	0.07	0.000	0.004	0.013	0.0001	0.000	0.001

PHYSICAL PROPERTIES

Ht Cd	Tensl	Yield	%Elong	%R.A.	HBW	Charpy	Ft-Lbs	Grain Sz	Lat Expsn	%-Shear	Test Temp
1F536N	73400	54100	35.40	75.58	149/149	TRANSV.	97/94/68		.049"/.053"/.042"	60/55/55	-50 F
1R231NM	75100	54400	30.80	73.56	163/163	TRANSVERSE	76/71/57		.050"/.046"/.036	45/40/35	-50 F
1F569H1	91000	72800	26.60	65.59	187/187	TRANSVERSE	76/83/86		.043"/.043"/.041"	50/50/45	-50 F

NOTES

Ht Cd	Country of Melt	Note
1F536N	United States	NORMALIZED, 1650F FOR 1HR/INCH, AIR COOL Fine Grain Practice
1R231NM	United States	NORMALIZED, 1650F FOR 1HR/INCH, AIR COOL Fine Grain Practice
1F569H1	United States	1650F 1HR/INCH, WATER QUENCH, TEMPER 1000F 1HR/INCH, AIR COOL Fine Grain Practice

All flanges bearing the "BOLTEX" name are machined in the United States at our Houston Texas facility. Flanges marked with "MADE IN USA" or "USA" are made from steel melted in the United States. Country of melt is shown in the notes section above. Billet used in the manufacturing process has been fully killed. Boltex flanges are manufactured, sampled, tested and inspected per MSS SP-44 2019 PSL1.

It is guaranteed that all Boltex forged steel flanges marked B16 or MSS, adhere to and comply with all dimensions and tolerances in accordance with specification ASME B16.36-2020, ASME B16.5-2020/ASME B16.47-2020 and MSS SP-44-2019 PSL1 as applicable for flanges. Boltex flanges have a surface finish of 125/250 roughness average (Ra) on all full face and raised face finishes.

We certify our flanges marked B16 or MSS, are capable of passing a hydrostatic test compatible with their rating and all test results and process information contained herein are correct and true as contained in company records. The flanges meet the requirements of the latest edition of NACE MR0175/ISO15156 and MR0103/ISO17945.

Our quality system has been registered by DNV-GL to ISO 9001, Certificate No.: 142240-2013-AQ-USA-ANAB. This MTR complies with EN 10204-3.1. Our flanges satisfy the material requirements for PED 2014/68/EU ANNEX 1, Section 4.3, Type 3.1. Notwithstanding the absence of a signature, the organization submitting either a printed certificate or an EDI transmitted certificate is responsible for report content or the report as per ASTM A961-21 Section 19.5.

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Quality Assurance Manager

APPROVED

Kevin.Jackson , 2/4/2026, 8:58:25 AM